ESENOTES.COM

STRUCTURE ANALYSIS

BY JASPAL SIR

TABLE OF CONTENTS

I	Structure Analysis: Introduction	1
II	Determinacy and Indeterminacy (Stability)	07
III	Trusses	84
IV	Influence Line Diagram	114
V	Arches	161
VI	Cables	197
VII	Analysis of Determinate Frames & Beams	204
VIII	Analysis of Indeterminate Structure	210
A	Force Method	214
	Method of Consistent Deformation	214
	Betti's Theorem	218
	Maxwell Reciprocal Theorem	220
	Castiglian's Theorem	220
	Strain Energy Method	221
	Three Moment Equation Method	232
B	Displacement Method	242
	Slope Deflection Method	242
	Moment Distribution Method	296
	Matrix Method of Analysis	341
	Column Analogy Method	364
	Kani's Method	369

ESENOTES.COM

Lesson 1 Feb 16

Structural Analysis

- structure can be defined as body of connected parts that is designed to carry loads even it is not intended to be occupied by us.

For eg: Bridges, dams, Railways, Retaining wall, tunnel, canals etc.

- The aim of structural analysis is to find force/moments in various components/parts of the structure.
- For the structure to remain in equilibrium, net forces (force & moments) acting on it must be zero in all directions.

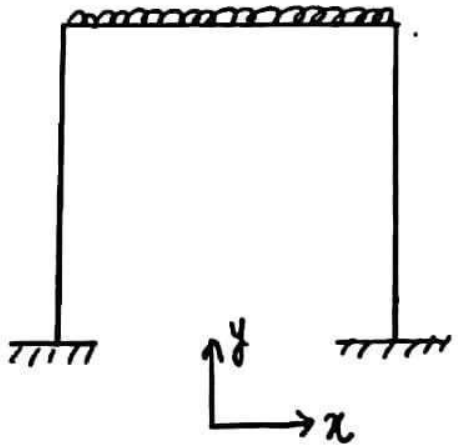
Note:→ If net force acting on body is zero & that body is at rest then it is termed to be in static equilibrium.

For eg:→ Aircraft flying, train running, vessel sailing, car moving with const speed : Equilibrium.
But, Bridge, Dam, canal, retaining wall : static Equilibrium

- In a 2D structure or planar structure (in which all the members & forces are in one plane only), the equations of equilibrium are

$$\left.\begin{aligned} \Sigma F_x &= 0 \\ \Sigma F_y &= 0 \\ \Sigma M_z &= 0 \end{aligned}\right\} \text{3no's}$$

The above structure is assumed to be in x-y plane.



written by Roopali
ESENOTES.COM

ESENOTES.COM

Lesson 23 Mar 11.

ARCHES

- An arch can be defined as two-dimensional structural element which is curved in elevation and is supported at ends by rigid or hinged supports which are capable of developing the desired thrust to resist the loads.

- It resist the external load through its profile
- Commonly used terms relating to arch are as follows :-

a) Springing

This is the point where the arch axis meets the supporting structure (column, pier, wall, abutments)

b) Crown:

The highest point on the curved axis of an arch.
If arch is symmetrical with springing ~~, it is~~ at same level it will lie above the mid point of the arch-span., else not.

c) Soffit

This is the lower surface of the arch which is normally curved in shape.

d) Rise

The vertical height of the crown above the springing is the rise of the arch.

e) Span

The horizontal distance b/w the springings is called span

Types of Arch

- Arches can be classified as follows

(A) On the basis of number of span.

(i) Simple Arch: It consist of single span structure.

(ii) Multiple Arch: It consists of multi-span structure

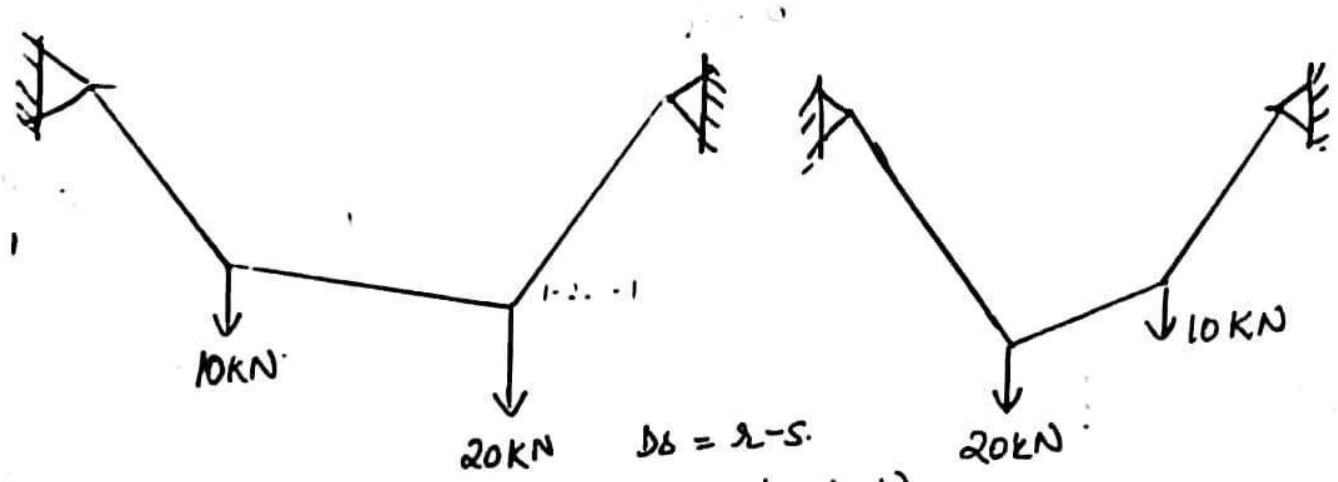
ESENOTES.COM

ESENOTES.COM

Lesson 27 Mar 15

CABLES

- Cables are tension members that cannot be extended axially.
- These are flexible in nature such that it changes its shape with different types & position of loads.
- These are not capable of resisting bending and shear



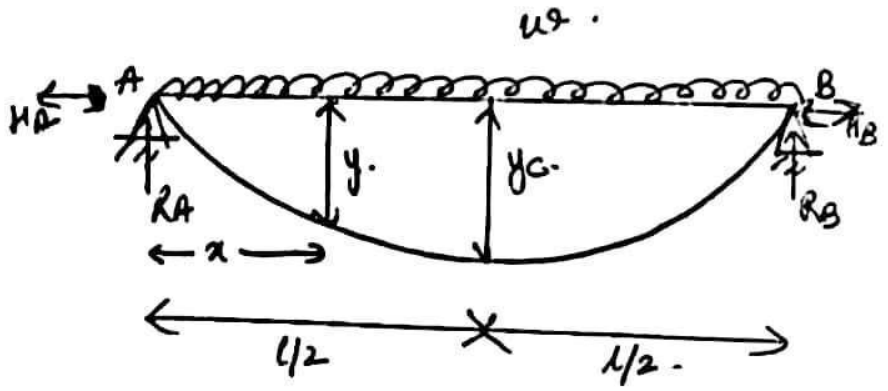
$Ds = r - s$
$= 4 - (3 + 1 - 1)$
$= -1$

Equation of Cable.

$\Sigma F_x = 0$
$H_A = H_B = H$

$\Sigma F_y = 0$
$R_A + R_B = wl$

$\Sigma M_B = 0$
$R_A \times l = wl \cdot \frac{l}{2}$
$R_A = R_B = \frac{wl}{2}$



y_c = sag of cable.

$M_C = 0 \rightarrow R_A \times \frac{l}{2} - w(l/2)(l/4) - Hy_c = 0$

ESENOTES.COM

ESENOTES.COM

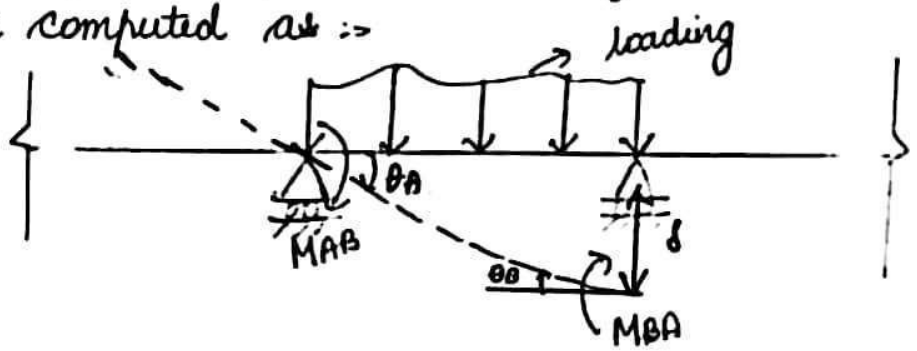
Displacement Method of Analysis $D_K < D_S$.

- In displacement method of analysis, joint displacement (θ, Δ) are taken as unknown.
- Here force displacement relationship is used & displacement are found using equilibrium equations.
- The displacements are then put back into the force displacement relationship & the member forces are computed
- It can be done by :-

(A) Slope deflection Method.

- It is used for analysis of Indeterminate structures like continuous beam & frames
- In this method unknowns are joint displacement i.e. slope & deflection. which is equal to degree of freedom (D_K)
- It forms the basis of Moment Distribution Method, Kani's method, stiffness matrix method.
- In slope deflection method, the force displacement equation is called as "slope deflection equation".

i) Slope deflection equation ~~can be~~ for continuous beam can be computed as :-



δ = deflection of Joint "B" w.r.t. joint "A".

θ_A & θ_B = Rotation of Joint A & B. resp.

M_{AB} & M_{BA} = internal member end moment at joint A & B

ESENOTES.COM